

Jervell and Lange-Nielsen Syndrome Type 1 (JLNS1): A Comprehensive Disease Characteristics Report

Disease: Jervell and Lange-Nielsen Syndrome 1 · **MONDO:** 0024540 · **OMIM:** #220400 · **Category:** Mendelian (autosomal recessive channelopathy)

Summary

Jervell and Lange-Nielsen syndrome type 1 (JLNS1) is a rare, severe **autosomal recessive cardio-auditory channelopathy** caused by **biallelic loss-of-function mutations in *KCNQ1*** (the α -subunit of the slow delayed-rectifier potassium current, I_{Ks}). *KCNQ1* mutations account for roughly **90% of JLNS**, with the remainder (JLNS2) caused by biallelic *KCNE1* mutations encoding the β -subunit. The syndrome is defined by the combination of **profound congenital bilateral sensorineural deafness** and **extreme QT-interval prolongation** (mean QTc ~557 ms; diagnostic threshold >500 ms) with a very high risk of syncope, torsades de pointes, and sudden cardiac death. Prevalence is estimated at 1/1,000,000 to 1/200,000.

The unifying mechanism is loss of the same I_{Ks} channel in two tissues. In **ventricular cardiomyocytes**, loss of I_{Ks} removes the repolarization reserve that shortens the action potential during sympathetic activation, producing marked QT prolongation and adrenergically-triggered arrhythmia. In the **cochlea**, the *KCNQ1/KCNE1* channel sits at the apical membrane of stria marginal cells and vestibular dark cells, where it secretes K^+ into the endolymph to complete the cochlear K^+ cycle and maintain the high- K^+ , +80 mV endocochlear potential essential for hearing; its loss collapses endolymph K^+ secretion and causes deafness. A *Kcnq1* knockout mouse faithfully recapitulates the inner-ear pathology. A third, under-recognized arm of the disease is **gastrointestinal**: the same channel recycles K^+ across the parietal-cell apical membrane, so its loss causes **hypochlorhydria, compensatory hypergastrinemia, gastric mucosal hyperplasia, and secondary iron-deficiency anemia**.

Clinically, JLNS1 is one of the most malignant forms of long QT syndrome: 86% of patients have cardiac events, half are symptomatic by age 3, and >25% die suddenly. **Beta-blockers are first-line but only partially effective** (51% event despite therapy), so high-risk patients (QTc >550 ms, syncope in the first year of life) require **left cardiac sympathetic denervation and/or an implantable cardioverter-defibrillator**. Deafness is managed with **cochlear implantation**, with auditory outcomes comparable to non-syndromic deafness provided strict peri-operative cardiac precautions are followed. Emerging **KCNQ1 suppression-and-replacement (SupRep) gene therapy** offers a mutation-independent, proof-of-concept correction strategy of particular relevance to JLNS1's biallelic-null genotype.

Key Findings

Finding 1 — JLNS1 is an autosomal recessive channelopathy caused by biallelic *KCNQ1* mutations (~90% of cases)

JLNS is caused by homozygous or compound heterozygous mutations in **KCNQ1 (JLNS1, ~90–90.5% of cases)** or **KCNE1 (JLNS2)**, encoding the α - and β -subunits, respectively, of the slow delayed-rectifier potassium current I_{Ks} . Inheritance is autosomal recessive; heterozygous carriers manifest the milder, dominantly-inherited Romano-Ward LQT1 phenotype. Prevalence is estimated between 1/1,000,000 and 1/200,000 worldwide. KCNE1-associated JLNS2 tends to follow a more benign clinical course than KCNQ1-associated JLNS1.

"Most mutations (90.5%) are on the KCNQ1 gene; mutations on the KCNE1 gene are associated with a more benign course." — [PMID: 17646758](#)

"Jervell and Lange-Nielsen syndrome (JLNS) is a rare but severe autosomal recessive disease characterized by profound congenital deafness and a prolonged QTc interval (greater than 500 milliseconds)... The prevalence of JLNS is about 1/1000000 to 1/200000 around the world." — [PMID: 32508908](#)

Ontology anchors: Gene *KCNQ1* (HGNC:6294), *KCNE1* (HGNC:6240); MONDO:0024540; OMIM #220400. GO:0005249 (voltage-gated potassium channel activity); GO:0086003 (cardiac muscle cell contraction); CHEBI:29103 (potassium ion, K⁺).

Finding 2 — JLNS is a severe, early-onset LQTS variant with a high cardiac-event rate and only partial beta-blocker efficacy

In the landmark international registry of 186 JLNS patients (Schwartz/Denjoy et al., 2006), **86% had cardiac events and 50% were symptomatic by age 3**, with markedly prolonged repolarization (**mean QTc 557 ± 65 ms**). Approximately **95% of arrhythmic events were triggered by emotion or exercise** (sympathetic activation), consistent with the LQT1 mechanism. **QTc >550 ms and a history of syncope in the first year of life are independent predictors of cardiac arrest/sudden death**. Beta-blockers, although first-line, provide only partial protection: 51% of patients had events despite therapy and 29% suffered cardiac arrest or sudden death. Females are at somewhat lower risk. More than 25% of JLNS patients die suddenly.

"Most patients (86%) had cardiac events and 50% were symptomatic already by age 3. Their QTc was markedly prolonged (557 ± 65 ms)." — [PMID: 17646758](#)

"A QTc>550 ms and history of syncope during the first year of life are independent predictors of subsequent CA/SD." — [PMID: 17646758](#)

"beta-blockers have only partial efficacy as 51% of the patients had events despite therapy and 29% had CA/SD." — [PMID: 17646758](#)

The nationwide Spanish paediatric LQTS registry (371 children) corroborates this risk profile: major arrhythmic events clustered in "high-risk genotypes" — explicitly including Jervell-Lange-Nielsen — and in children with QTc ≥ 550 ms or very early (fetal/neonatal) presentation ([PMID: 42466893](#)).

Ontology anchors: HP:0001657 (Prolonged QT interval); HP:0004308 (Ventricular arrhythmia / torsade de pointes); HP:0001279 (Syncope); HP:0001645 (Sudden cardiac death); HP:0001695 (Cardiac arrest).

Finding 3 — Mechanism: loss of KCNQ1/KCNE1 I_{Ks} abolishes marginal-cell K^+ secretion into endolymph, causing deafness

The KCNQ1/KCNE1 channel is expressed at the **apical membrane of strial marginal cells in the cochlea and vestibular dark cells**, where it secretes K^+ into the endolymph. This step completes the **cochlear K^+ recycling cycle** required to maintain the high- K^+ , +80 mV endolymph and endocochlear potential (EP) that drives sensory hair-cell

mechanotransduction. An additional KCNQ1/KCNE1-mediated K⁺-diffusion potential contributes to EP formation within the stria vascularis. Loss of this channel collapses endolymph K⁺ secretion, producing **profound congenital sensorineural deafness**.

"Deafness results also from mutations of KCNQ1 or KCNE1, subunits of a K(+) channel that carries K(+) from strial marginal cells and vestibular dark cells into endolymph." — PMID: 12097719

"An additional K(+)-diffusion potential formed by KCNQ1/KCNE1-K" — PMID: 20012478

The detailed K⁺ cycle (from PMID: 12031509) situates KCNQ1/KCNE1 as the **final apical secretory step**: K⁺ enters the strial marginal cell basolaterally via SLC12A2 (NKCC1) and Na⁺/K⁺-ATPase, then exits apically into endolymph via KCNQ1/KCNE1, "which concludes the cochlear cycle." The same architecture operates in the vestibular labyrinth via dark cells, explaining associated vestibular dysfunction.

Ontology anchors: UBERON:0002215 (stria vascularis); UBERON:0002227 (cochlea); UBERON:0007499 (endolymph); CL:0000710 (marginal cell of stria vascularis); GO:0071805 (potassium ion transmembrane transport); GO:0042472 (inner ear morphogenesis). HP:0000407 (Sensorineural hearing impairment); HP:0000365 (Hearing impairment).

Finding 4 — The *Kcnq1* knockout mouse recapitulates JLNS inner-ear pathology

Kcnq1 (Kvlqt1) knockout mice are **deaf and display circling behavior** (vestibular dysfunction). Temporal-bone histopathology reveals **marked atrophy of the stria vascularis, contraction/collapse of the endolymphatic compartments, complete degeneration of the organ of Corti, and secondary spiral-ganglion degeneration** — findings that parallel human JLNS temporal-bone pathology and reflect disturbed endolymph production by the stria vascularis.

"Kcnq1 knockout mice were deaf and demonstrated circling behavior. They exhibited a marked atrophy of the stria vascularis, contraction of the endolymphatic compartments, and collapse and adhesion of surrounding membranes. There was a complete degeneration of the organ of Corti and an associated degeneration of the spiral ganglion." — PMID: 15891643

Ontology anchors: NCBITaxon:10090 (*Mus musculus*); model gene *Kcnq1* (NCBI Gene ID 16535). UBERON:0002365 (organ of Corti); UBERON:0002228 (spiral ganglion).

Finding 5 — Management combines cardiac risk reduction with cochlear implantation

Cardiac management centers on **beta-blockers as first-line therapy** (partial protection only), escalating in high-risk patients (QTc >550 ms, syncope in the first year, aborted arrest) to **ICD implantation and/or left cardiac sympathetic denervation (LCSD)**, alongside strict avoidance of QT-prolonging drugs and sympathetic triggers. Atrial pacing combined with beta-blockade has been proposed for selected young high-risk patients ([PMID: 27451284](#)). **Cochlear implantation** is the standard hearing rehabilitation and yields auditory/language outcomes **not worse than in non-syndromic sensorineural deafness**, provided cardiac events are controlled and strict peri-operative precautions are used (beta-blockade, avoidance of sympathetic stimulation and QT-prolonging drugs, defibrillator standby, magnesium). Importantly, LQTS/JLNS syncope can be **misdiagnosed as epilepsy**, delaying correct therapy.

"Fatal Arrhythmias were encountered intra-operatively in five patients which was treated with cardiac pacing." — [PMID: 31846911](#)

"The auditory and language outcome after cochlear implantation in this syndrome is not worse than those in patients with non-syndromic sensorineural deafness." — [PMID: 18805595](#)

"Current treatment strategies include high doses of beta-blocker medication, left cardiac sympathetic denervation, and ICD placement, which is challenging in young children." — [PMID: 27451284](#)

Ontology anchors (NCIT): Beta-adrenergic blocker therapy (propranolol, nadolol); Implantable cardioverter-defibrillator; Left cardiac sympathetic denervation (sympathectomy); Cochlear implantation.

Finding 6 — Heterozygous carriers have milder LQT1; a common founder allele (R518X) exists

Heterozygous carriers of JLNS-causing *KCNQ1* variants show a substantially milder phenotype than homozygous JLNS patients, illustrating a **gene-dosage effect**. In a genotype-stratified cohort, mean QTc was **551 ± 54 ms in JLNS homozygotes vs 441 ± 32 ms in heterozygous-**

JLNS carriers and 467 ± 36 ms in heterozygous non-JLNS carriers. The recessive JLNS phenotype (deafness + long QTc + ~47% cumulative aborted cardiac arrest) contrasts starkly with the benign dominant LQT1 phenotype of heterozygotes (QTc ~462 ms, ~1% arrest). The nonsense variant **p.R518X (c.1552C>T) in *KCNQ1*** is a common founder allele that causes JLNS when biallelic and LQT1 when heterozygous; in Sweden it is an old founder mutation (~28 generations) with heterozygote prevalence ~1:2,000–4,000, clustering geographically. **Consanguinity** contributes to JLNS in inbred families.

"the mean QTc was 551 ± 54 ms for JLNS, 441 ± 32 ms for HTZ-JLNS, and 467 ± 36 ms for HTZ-Non-JLNS" — [PMID: 38825991](#)

"The R518X/*KCNQ1* mutation is a common cause of autosomal recessive (Jervell and Lange Nielsen Syndrome- JLNS) and autosomal dominant long QT syndrome (LQTS) worldwide." — [PMID: 24552659](#)

"Clinical phenotype ranged from expectedly severe in JLNS to surprisingly benign in LQTS (QTc 576 ± 61 ms vs. 462 ± 34 ms, cumulative incidence of (aborted) cardiac arrest 47% vs. 1%" — [PMID: 24552659](#)

Variant nomenclature: *KCNQ1* p.Arg518* (R518X), c.1552C>T — nonsense (loss-of-function), Pathogenic. At least one highly inbred family displayed apparent pseudo-dominant transmission with a milder phenotype and hearing/QT severity correlating ([PMID: 23668803](#)).

Finding 7 — JLNS has an extra-cardiac gastrointestinal phenotype: hypochlorhydria, hypergastrinemia, and iron-deficiency anemia

In a Swedish cohort of 14 patients with biallelic *KCNQ1* mutations, **a history of iron-deficiency anemia was present in 12/14 and subjective GI symptoms in 13/14**. Objective findings included elevated gastrin (7/9), elevated pepsinogen (6/7), elevated fecal calprotectin (9/9), and gastric mucosal hyperplasia/dysplasia (3/5 endoscoped). Elevated gastrin correlated significantly with iron deficiency/anemia ($p = 0.039$). ***KCNQ1* is essential for gastric acid secretion**, recycling K^+ across the parietal-cell apical membrane to supply the H^+/K^+ -ATPase; its loss causes **hypochlorhydria → impaired dietary iron absorption → iron-deficiency anemia**, with compensatory **hypergastrinemia** that constitutes a potential long-term gastric-cancer risk factor.

"A history of iron-deficiency anaemia in 12 of 14 patients and subjective gastrointestinal symptoms in 13 of 14 patients was revealed." — [PMID: 22805636](#)

"We propose that the Jervell and Lange-Nielsen Syndrome phenotypically includes gastrointestinal symptoms/signs and secondary iron-deficiency anaemia owing to hypochlorhydria on the basis of KCNQ1 mutations. The resultant elevated gastrin level is a potential risk factor for later gastrointestinal cancer." — [PMID: 22805636](#)

Ontology anchors: HP:0001891 (Iron deficiency anemia); HP:0011473 (Achlorhydria); HP:0003637 (Hypergastrinemia); UBERON:0000945 (stomach); CL:0000162 (parietal cell); GO:0030007 (intracellular potassium ion homeostasis); CHEBI:29103 (potassium ion). Body system: digestive/gastrointestinal (UBERON:0001007).

Finding 8 — I_{Ks} is a multi-subunit regulated complex; patient iPSC-cardiomyocytes recapitulate LQT1 and reveal therapeutic limits for null- I_{Ks} JLNS

The I_{Ks} channel complex is built from **Kv7.1 (KCNQ1) co-assembled with the β -subunit KCNE1**, plus accessory regulators **PIP2, calmodulin, and yotiao (AKAP9)**. This regulatory flexibility lets one channel serve both cardiac repolarization and endolymph K^+ maintenance. Patient-derived **iPSC-cardiomyocytes** from LQT1/KCNQ1 carriers reproduce the disease: markedly **prolonged action-potential duration (APD)** due to reduced KCNQ1-mediated I_{Ks} , with some mutations (e.g., R190Q, M437V) acting via dominant-negative trafficking defects. The **NOS1AP modifier locus** interacts with prolonged repolarization to modulate arrhythmia risk and penetrance. Critically, **I_{Ks} -activator compounds shorten APD in LQT2 but not LQT1**, because LQT1/JLNS I_{Ks} is intrinsically impaired — implying that channel-activator strategies have limited utility when I_{Ks} is **biallelically null**, as in JLNS1.

"The I_{Ks} channel complex is formed by the co-assembly of Kv7.1 (KCNQ1), a voltage-gated potassium channel, with its β -subunit, KCNE1 and the association of numerous accessory regulatory molecules such as PIP2, calmodulin, and yotiao." — [PMID: 32581825](#)

"The duration of the action potential was markedly prolonged in 'ventricular' and 'atrial' cells derived from patients with long-QT syndrome type 1, as compared with cells from control subjects." — [PMID: 20660394](#)

"DHA-gly shortened APD in vitro and QT interval ex vivo in WT and LQT2 rabbits, but not in LQT1" — [PMID: 40794559](#)

Additional supporting model data: iPSC-CM from an M437V *KCNQ1* patient exhibited APD prolongation from reduced I_{Ks} ([PMID: 31245483](#)); human iPSC-CM engineered heart tissue has a low repolarization reserve making it a sensitive QT/arrhythmia test system ([PMID: 29925535](#)); *NOS1AP* minor alleles associate with NOS1 loss-of-function that aggravates the arrhythmia phenotype in prolonged-repolarization syndromes ([PMID: 32061134](#)).

Ontology anchors: GO:0086089 (voltage-gated potassium channel activity involved in cardiac repolarization); GO:1990573 (potassium ion import across plasma membrane); regulators AKAP9/yotiao, CALM1 (calmodulin); PIP2 (phosphatidylinositol 4,5-bisphosphate). Model: hiPSC-cardiomyocyte.

Finding 9 — KCNQ1 suppression-and-replacement (SupRep) gene therapy is a mutation-independent proof-of-concept for LQT1/JLNS1

A dual-component **KCNQ1 "suppression-and-replacement" (SupRep)** gene therapy achieves **mutation-independent suppression of endogenous KCNQ1** (via shRNA) combined with delivery of an **shRNA-immune replacement KCNQ1 cDNA**. In TSA201 cells and LQT1 patient iPSC-cardiomyocytes, it **corrected the prolonged action-potential/QT phenotype** — the first proof-of-principle gene therapy for complete correction of long QT syndrome type 1. Because it is mutation-independent, a single construct could in principle treat any *KCNQ1* genotype, including the biallelic null alleles of JLNS1 — an important conceptual advance, since channel-activator drugs cannot rescue a null channel.

"This study provides the first proof-of-principle gene therapy for complete correction of long QT syndrome." — [PMID: 33504163](#)

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

Biallelic loss-of-function mutation in KCNQ1 (e.g., R518X; homozygous/compound het)

| leads to



Absent or non-functional Kv7.1 α -subunit → failure to form functional KCNQ1/KCNE1 IKs channel

| results in



Loss of the slow delayed-rectifier K⁺ current (IKs) in multiple K⁺-transporting epithelia/myocytes



—[CARDIAC BRANCH]—

Loss of IKs removes repolarization reserve in ventricular cardiomyocytes → prolonged action-potential duration → QT prolongation (QTc ~557 ms). Under sympathetic activation (emotion/exercise, ~95% of triggers), inadequate IKs upregulation → early afterdepolarizations → torsades de pointes → syncope / cardiac arrest / sudden death (>25%).



—[COCHLEAR BRANCH]—

Loss of apical KCNQ1/KCNE1 in strial marginal cells & vestibular dark cells → failure of K⁺ secretion into endolymph → collapse of K⁺-rich endolymph & +80 mV endocochlear potential → loss of hair-cell mechanotransduction driving force → (developmentally) strial atrophy, endolymphatic collapse, organ of Corti degeneration → profound congenital deafness (+ vestibular dysfunction).



—[GASTRIC BRANCH]—

Loss of parietal-cell apical KCNQ1 K⁺ recycling → impaired H⁺/K⁺-ATPase-driven acid secretion → hypochlorhydria →
(a) impaired iron absorption → iron-deficiency anemia
(b) compensatory hypergastrinemia → gastric mucosal hyperplasia/dysplasia → potential long-term cancer risk.

The elegance of JLNS1's pathophysiology is that a **single molecular lesion** — loss of one K⁺ channel — produces a multi-organ syndrome because **KCNQ1/KCNE1 performs the same biophysical job (K⁺ efflux/recycling) in three different epithelia**. Upstream is the genetic lesion and channel loss; downstream are the tissue-specific consequences. The cardiac and cochlear arms are the classic defining dyad; the gastric arm is a more recently characterized, under-recognized systemic extension. The cardiac arm is specifically an **adrenergic/gene-environment phenotype**: I_{Ks} is the current the healthy heart recruits (via β -adrenergic/PKA

signaling anchored by yotiao/AKAP9) to shorten repolarization during sympathetic drive; without it, sympathetic stress becomes dangerous, which is why ~95% of events are triggered by emotion or exercise.

Comparison of phenotypic arms

Tissue / cell type	Normal KCNQ1/ KCNE1 role	Consequence of loss	Clinical manifestation
Ventricular cardiomyocyte	I_{Ks} repolarization reserve, esp. under β -adrenergic drive	Prolonged APD/QT, EADs	QTc ~557 ms, torsades, syncope, SCD
Strial marginal cell (UBERON:0002215) / vestibular dark cell	Apical K^+ secretion into endolymph; EP maintenance	Endolymph K^+ / EP collapse; strial atrophy	Profound congenital sensorineural deafness; vestibular dysfunction
Gastric parietal cell (CL:0000162)	Apical K^+ recycling for H^+/K^+ -ATPase	Hypochlorhydria; hypergastrinemia	Iron-deficiency anemia; gastric hyperplasia

Genotype–phenotype and dosage

Genotype	Representative QTc	Deafness	Cardiac-arrest risk
Biallelic <i>KCNQ1</i> LOF (JLNS1)	~551–576 ms	Profound, congenital	~47% cumulative
Heterozygous JLNS-allele carrier	~441 ms	Normal hearing	~1%
Heterozygous non-JLNS LQT1 allele	~467 ms	Normal hearing	Low
Biallelic <i>KCNE1</i> (JLNS2)	Prolonged	Present	More benign than JLNS1

Section-by-Section Reference Compilation

1. Disease Information

JLNS1 is an autosomal recessive cardio-auditory syndrome (deafness + long QT). Identifiers: **MONDO:0024540**; **OMIM #220400**; **Orphanet ORPHA:90647**; ICD-10 **I45.81** (long QT syndrome); MeSH **D029593** (Jervell-Lange Nielsen Syndrome). Synonyms: Jervell and Lange-Nielsen syndrome; cardioauditory syndrome of Jervell and Lange-Nielsen; surdo-cardiac syndrome; long QT syndrome with deafness. Information is derived predominantly from **aggregated disease-level resources** (international LQTS registries, OMIM, Orphanet) and case series rather than individual EHR.

2. Etiology

Primary cause is genetic — biallelic LOF mutations in *KCNQ1* (JLNS1) or *KCNE1* (JLNS2) (Findings 1, 6). No infectious or purely environmental cause. **Genetic risk factors:** the causal variants; **founder alleles** (R518X) and **consanguinity** increase incidence in specific populations. **Modifiers:** *NOS1AP* polymorphisms interact with prolonged repolarization to modulate arrhythmia penetrance ([PMID: 32061134](#)). **Acquired arrhythmia multipliers (triggers, not disease cause):** QT-prolonging drugs, sympathetic activation (emotion, exercise, sudden auditory stimuli), electrolyte disturbance. **Protective factors:** female sex confers lower cardiac risk; strict trigger avoidance and beta-blockade reduce events. **Gene-environment interaction:** the phenotype is fundamentally adrenergic — I_{Ks} loss becomes clinically dangerous under sympathetic stress (95% of events triggered by emotion/exercise).

3. Phenotypes

Core phenotypes: **profound bilateral congenital sensorineural deafness** (HP:0000407; essentially fully penetrant, congenital, stable, severe) and **prolonged QT interval / torsades / syncope / sudden death** (HP:0001657, HP:0004308, HP:0001279, HP:0001645; onset neonatal-early childhood, 50% symptomatic by age 3, severe, episodic-triggered). Extra-cardiac: **iron-deficiency anemia** (HP:0001891, ~85%), **hypochlorhydria/hypergastrinemia**, GI symptoms; **vestibular dysfunction** (delayed motor milestones from balance impairment). Quality-of-life impact is substantial: lifelong deafness plus activity restriction (avoidance of competitive sport/emotional stress), device dependence (ICD/cochlear implant), and the psychological burden of sudden-death risk.

4. Genetic / Molecular Information

Causal genes: *KCNQ1* (11p15.5–15.4; OMIM 607542; HGNC:6294) — ~90%; *KCNE1* (21q22.12; OMIM 176261; HGNC:6240) — JLNS2. **Variant types:** nonsense (e.g., R518X), missense, frameshift, splice-site — predominantly loss-of-function; JLNS requires biallelic hits. Functional consequence: **loss of function / loss of I_{Ks}** ; some heterozygous LQT1 alleles act dominant-negative (trafficking defects, e.g., R190Q, M437V). **Origin:** germline. **Allele frequency:** individual pathogenic alleles are rare in gnomAD; founder alleles are regionally enriched. **Modifier:** *NOS1AP*. Epigenetic/chromosomal contributions: none established.

5. Environmental Information

No toxin, radiation, pollution, occupational, or infectious cause. Relevant **acquired arrhythmia modifiers:** QT-prolonging medications, hypokalemia/hypomagnesemia, sympathetic triggers. Lifestyle: avoidance of strenuous/competitive exercise and emotional stressors is recommended.

6. Mechanism / Pathophysiology

See the ordered causal chain and Mechanistic Model above (Findings 3, 4, 7, 8). Molecular pathway: β -adrenergic (PKA/cAMP) $\rightarrow$ yotiao/AKAP9-anchored phosphorylation normally augments I_{Ks} ; channel loss removes this adaptive repolarization response. Cellular processes: ion transport, membrane repolarization, endolymph homeostasis, acid secretion. Protein dysfunction: loss/trafficking failure of Kv7.1. Cell types: ventricular cardiomyocytes, stria marginal cells, vestibular dark cells, gastric parietal cells.

7. Anatomical Structures Affected

Primary organs: heart (ventricular myocardium/conduction — UBERON:0000948, UBERON:0002349) and inner ear (cochlea/stria vascularis UBERON:0002227/0002215; vestibular labyrinth). **Secondary:** stomach (UBERON:0000945), hematologic system (secondary anemia), brain (secondary hypoxic-ischemic injury from arrests). **Body systems:** cardiovascular, nervous/sensory (auditory + vestibular), digestive. **Subcellular:** apical plasma membrane of epithelial cells (GO:0016324); cardiomyocyte sarcolemma. **Lateralization:** bilateral (hearing) and systemic (cardiac).

8. Temporal Development

Onset: deafness is **congenital**; cardiac events begin in infancy/early childhood (50% symptomatic by age 3; fetal/neonatal bradycardia can be a harbinger of malignant genotype). **Course:** deafness is stable/non-progressive; the cardiac phenotype is **lifelong and episodic**, with arrhythmic events triggered by adrenergic stress. **Critical period:** first year of life — syncope then is a strong predictor of subsequent cardiac arrest/sudden death. Untreated natural history carries very high early mortality.

9. Inheritance and Population

Autosomal recessive. Prevalence 1/1,000,000–1/200,000. **Penetrance** of deafness essentially complete in biallelic patients; cardiac severity variable/expressivity variable. **Founder effects** (R518X in Sweden, ~28 generations, carrier ~1:2,000–4,000) and **consanguinity** raise regional incidence. School-based ECG screening of deaf children found JLNS in ~0.9% ([PMID: 34308870](#)). Sex ratio ~equal for the syndrome, but **females have lower cardiac-event risk**.

10. Diagnostics

ECG (Bazett QTc; diagnostic >500 ms, often >550) with exercise stress testing; **audiometry** (profound bilateral SNHL). **Genetic testing** is confirmatory: single-gene *KCNQ1/KCNE1* testing, LQTS/hearing-loss gene panels, or WES/WGS. Genetic testing is essential to distinguish true JLNS from the coincidental co-occurrence of independent deafness (e.g., *PCDH15*) and LQT genes ([PMID: 41563185](#)). Supportive: elevated gastrin/pepsinogen/calprotectin and iron studies for the GI arm. **Differential diagnosis:** Romano-Ward LQTS with unrelated deafness, Pendred syndrome (deafness+goiter), Usher/Waardenburg (deafness without LQT), other LQTS genotypes; seizure disorders (syncope misdiagnosed as epilepsy).

11. Outcome / Prognosis

Malignant without treatment: >25% sudden death; 86% cardiac-event rate; 29% cardiac arrest/SD even on beta-blockers. **Prognostic factors:** QTc >550 ms, syncope in first year, aborted arrest, male sex, high-risk genotype = worse; female sex, controlled triggers, LCSD/ICD = better. Deafness causes permanent disability mitigated by cochlear implantation. With comprehensive management (beta-blockers + LCSD/ICD + implant), survival and function improve substantially.

12. Treatment

Pharmacotherapy: beta-blockers (propranolol, nadolol) first-line (NCIT: beta-adrenergic blocker therapy); mexiletine adjunct in some LQTS. **Device/interventional:** ICD; left cardiac sympathetic denervation (thoracoscopic — 73% arrhythmia reduction in ventricular-arrhythmia cohorts, [PMID: 24268954](#)); atrial pacing + beta-blockade in selected young patients; catheter ablation of triggering PVCs in refractory storm ([PMID: 33040543](#)). **Deafness:** cochlear implantation. **Supportive:** iron supplementation for the anemia arm. **Experimental/advanced:** KCNQ1 SupRep gene therapy (mutation-independent proof-of-concept, [PMID: 33504163](#)); note I_{Ks} -activator drugs fail in LQT1/null- I_{Ks} ([PMID: 40794559](#)).

13. Prevention

No primary prevention of the genetic disease. **Genetic counseling** and **carrier/cascade screening** in affected families; prenatal/preimplantation testing feasible. **Secondary prevention:** ECG screening of congenitally deaf children detects unrecognized JLNS ([PMID: 34308870](#)). **Tertiary prevention:** trigger avoidance (QT-prolonging drugs, competitive sport, sudden loud stimuli), beta-blockade, device therapy, and peri-operative arrhythmia precautions during cochlear implantation.

14. Other Species / Natural Disease

Orthologous gene *Kcnq1* in mouse (NCBI Gene 16535, NCBITaxon:10090) and rat. No prominent naturally-occurring companion-animal JLNS equivalent is established; the disease is studied primarily via engineered rodent models. The mechanism (KCNQ1-dependent endolymph K^+ secretion and cardiac repolarization) is **evolutionarily conserved** across mammals.

15. Model Organisms

Mouse *Kcnq1* knockout — the key model, reproducing deafness, circling/vestibular dysfunction, striaal atrophy, and organ-of-Corti degeneration (Finding 4; [PMID: 15891643](#)); species differences in ventricular repolarization reserve limit full recapitulation of the human QT phenotype. **Human iPSC-derived cardiomyocytes / engineered heart tissue** from LQT1/*KCNQ1* patients recapitulate APD/QT prolongation and serve as platforms for mechanism and therapy testing (Findings 8, 9; [PMID: 20660394](#), [PMID: 31245483](#), [PMID: 29925535](#)). **Guinea-pig cardiomyocyte** pharmacological LQT1 models are used for modifier (NOS1AP) studies ([PMID: 32061134](#)). **Rabbit LQT1 models** test I_{Ks} -activator pharmacology ([PMID: 40794559](#)).

Evidence Base

PMID	Role in this report	Evidence type
17646758	Natural history: 90.5% KCNQ1, QTc 557 ms, 86% events, partial beta-blocker efficacy, risk predictors	Human registry (186 pts)
32508908	Definition, AR inheritance, prevalence, QTc>500 ms	Human case + review
12097719	KCNQ1/KCNE1 K ⁺ transport into endolymph by marginal/dark cells	Review/mechanism
20012478	KCNQ1/KCNE1 contributes to endocochlear potential	Mechanism
12031509	Full cochlear/vestibular K ⁺ cycle; KCNQ1/KCNE1 apical secretory step	Review
15891643	Kcnq1 KO mouse recapitulates inner-ear pathology	Mouse model
22805636	GI phenotype: hypochlorhydria, hypergastrinemia, iron-deficiency anemia	Human cohort (14 pts)
24552659	R518X founder allele; severe recessive vs benign dominant phenotype	Human genealogy/genetics
38825991	Gene-dosage QTc stratification	Human cohort
31846911	Peri-operative arrhythmia risk in cochlear implantation	Human case series
18805595	Cochlear implant outcomes equal to non-syndromic deafness	Human review
27451284	Cardiac treatment options; atrial pacing + beta-blockade	Human
32581825	I _{Ks} complex composition (KCNQ1/KCNE1/PIP2/CaM/yotiao)	Review
20660394	Patient iPSC-CM recapitulate APD prolongation	In vitro
31245483	M437V KCNQ1 iPSC-CM: reduced I _{Ks} , prolonged APD	In vitro
29925535	hiPSC-CM engineered heart tissue as QT/arrhythmia test system	In vitro
40794559	I _{Ks} -activator fails in LQT1	Rabbit/in vitro
33504163	KCNQ1-SupRep gene therapy proof-of-concept	In vitro/iPSC-CM
32061134	NOS1AP modifier of arrhythmia penetrance	GP-CM/iPSC-CM
42466893	JLNS as high-risk genotype in paediatric LQTS registry	

PMID	Role in this report	Evidence type
		Human registry (371)
34308870	JLNS prevalence (~0.9%) in deaf-school ECG screening	Human screening
41563185	Genetic testing distinguishes true JLNS from coincidental deafness+LQT	Human case
24268954	Thoracoscopic LCSD reduces ventricular arrhythmia burden	Human series
33040543	Ablation of triggering PVC in JLNS arrhythmia storm	Human case
23668803	Inbred family; hearing/QT severity correlation; consanguinity	Human family study

Limitations and Knowledge Gaps

- Rarity limits epidemiology.** Prevalence estimates span a five-fold range (1/1,000,000–1/200,000); incidence and geographic/variant distribution data are sparse outside founder populations.
- GI phenotype based on a single small cohort.** The hypochlorhydria/anemia/hypergastrinemia arm rests largely on 14 Swedish patients ([PMID: 22805636](#)); the proposed long-term gastric-cancer risk is inferred, not demonstrated longitudinally.
- Model species differences.** Mice have different ventricular repolarization reserve, so the *Kcnq1* KO reproduces deafness far better than the human QT/arrhythmia phenotype; iPSC-CMs model the cardiac arm but not the intact organ or the auditory/gastric arms.
- Therapeutic gap for null I_{Ks} .** Beta-blockers are only partially effective and I_{Ks} -activators cannot rescue biallelic-null channels; SupRep gene therapy remains pre-clinical (in vitro/iPSC-CM only), with no in vivo JLNS1 efficacy data.
- Modifier landscape incomplete.** Beyond *NOS1AP*, the genetic/environmental determinants of the wide cardiac-severity variability are not fully mapped.
- Abstract availability.** Several key papers (e.g.,

P 38825991

P 27451284

P 33504163

P 31846911

had limited/no abstract text available; claims rely on the verified snippets recorded during the investigation.

Proposed Follow-up Experiments / Actions

1. **Advance KCNQ1-SupRep toward in vivo proof of concept** in a humanized/biallelic-null JLNS1 model (AAV delivery to cardiomyocytes; evaluate QT correction and safety), given that activator drugs cannot rescue null I_{Ks} .
2. **Longitudinal GI surveillance cohort** to quantify hypochlorhydria, hypergastrinemia, iron-deficiency anemia frequency, and gastric-neoplasia risk across JLNS1 patients internationally; test iron/gastrin as biomarkers.
3. **Multi-organ JLNS1 modeling** — inner-ear-specific vs cardiac-specific conditional *Kcnq1* knockouts, plus iPSC-derived otic and gastric organoids — to dissect tissue-specific consequences and test correction per arm.
4. **Genotype–phenotype and modifier studies** in an expanded JLNS registry to identify modifiers (beyond *NOS1AP*) explaining cardiac-severity variability and refine risk stratification.
5. **Prospective ECG screening programs** in congenitally deaf children (building on [PMID: 34308870](#)) to reduce first-event sudden death through early diagnosis and beta-blockade.
6. **Peri-operative protocol standardization** for cochlear implantation in LQTS/JLNS (beta-blockade, magnesium, defibrillator standby, pacing readiness), validated across centers.

Report compiled from an autonomous 5-iteration literature-based investigation. Evidence types are labeled human clinical, model organism, or in vitro throughout. All quoted snippets were verified against source abstracts where available.

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